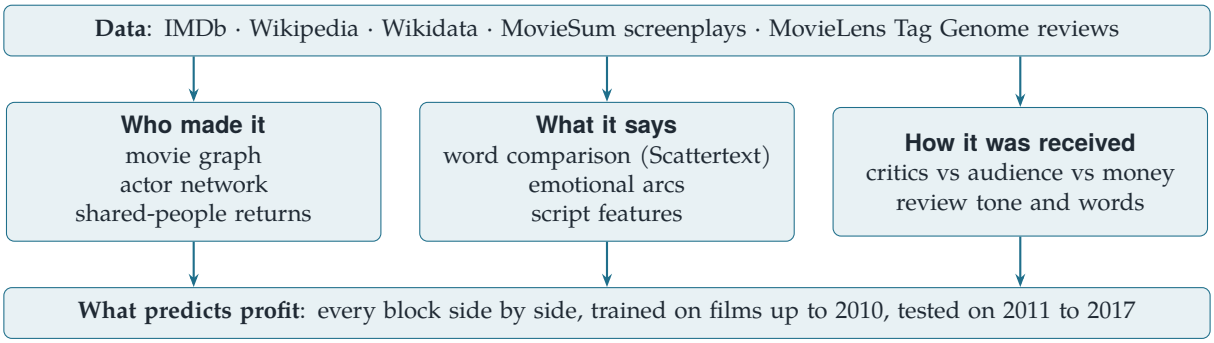
What makes a movie profitable? Project overview

Log-Log Legends: Àngela Buxó, Gyula Kürthy, Niklas Johansen · DTU 02805, autumn 2026 · state on 14 November · paper due 27 November, presentation 2 December · more detail in project-details.pdf · GitHub repo

Question. What makes a movie highly profitable, and does making money agree with being liked?

Profit is ln(worldwide gross/budget), "the multiple". It leaves out marketing and the cinemas' share, so it's a return on the reported budget, not real profit. **Sample:** 2,446 clean films with 100,000+ IMDb votes, or 6,851 with 10,000+.

Three strands, one model



What we've found

Who made it	The best predictor known before release. Studio and people track records lift out-of-sample R^2 from -0.01 to 0.20. Films that share people have similar returns.
What it says	Almost nothing once budget, genre and era are fixed. The word difference vanishes without franchise films, and the arcs show no clear link (one near miss).
How it was received	Money and liking agree only moderately (ρ 0.27 to 0.33). Reviews add to the model (R^2 0.24 to 0.31), but they come after release.

Possible one story: who makes a film predicts its return, what's in the script doesn't, and audiences only partly reward what earns.

Rules every strand follows

- Network measures use only earlier films.
- Every result gets a check: a shuffle null, a second method, or hand labels.
- Budget, era and genre are controlled, and franchise films get a separate check.
- No script text, review text or IMDb fields go into git.

Open

- Agree on the one story for the paper.
- Fill the human validation sheets (Scattertext, arcs, reviews).
- Merge Àngela's reviews branch and Niklas's actor-network branch.
- Arcs Step 11 (script tone vs review tone) needs the group to decide whether whole-review VADER counts as the same instrument.